

Introduction

Categorical variables are widely found in practical data sets and need to be transformed into numeric form for machine learning models to be utilized. Adult Income data set has several categorical variables including the high cardinality variable, 'native-country'. Using one-hot encoding on all categorical variables leads to increased dimensionality. This study aims to determine if frequency encoding of the high cardinality variable will decrease dimensionality while sustaining classification accuracy.

Method

In this experiment, the Adult Income dataset is used. Three encoding techniques have been compared. In the baseline technique, full one-hot encoding is used for all categorical features. In the second technique, rare category grouping is done first, where infrequent categories are collapsed to an "Other" category, and one-hot encoding is performed. In the third technique, the low cardinality features are one-hot encoded, whereas the high cardinality feature, native-country, is frequency encoded. A logistic regression algorithm is used as the classifier, and 5 fold cross-validation with accuracy as the evaluation metric is performed.

Results

Comparisons were made in terms of the dimensionality of generated features and cross validated accuracies of the three encoding techniques. Full one-hot encoding results in higher dimensions compared to the other two methods. This is because in rare category grouping, some categories are grouped together, while in hybrid encoding, a feature of high cardinality is replaced by frequency encoding to reduce dimensionality. Now that the dimensionality has been discussed, it can be investigated if reduction in dimensionality affects the accuracies of the models.

Results

The number of generated features as well as the cross-validation accuracy scores of the three encoding techniques were compared. One-hot encoding yielded the largest number of features among all encoding techniques tested. The rare category grouping technique was able to reduce the number of features through groupings of rarely occurring features, whereas the hybrid approach was able to achieve further dimensionality reduction by substituting the high cardinality variable with a frequency encoded feature.

Conclusion

In conclusion, the experiment shows that there are other forms of encoding that can help solve the issue of one-hot encoding in terms of dimensionality reduction. Rare category grouping solves this problem by reducing the number of columns, whereas hybrid encoding reduces the dimensionality further through frequency encoding for high cardinality variables. However, the findings of the experiment show that these two methods of encoding may be able to achieve accuracy equal to that achieved through complete one-hot encoding without the latter's drawbacks. Future research might consider more sophisticated types of encoding, such as target encoding, using different machine learning algorithms, including Random Forests and gradient boosting algorithms.